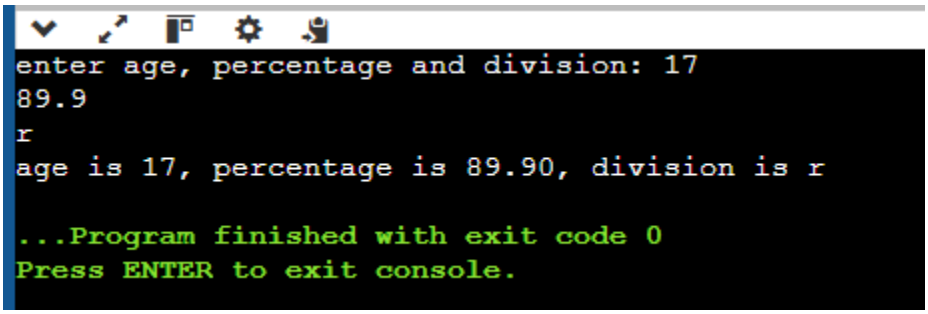
Laboratory Tasks

Task 1 : Different Data Types

```c
#include <stdio.h>
int main () {
    int a;
    float b;
    char c;
    printf("enter age, percentage and division: ");
    scanf("%d%f %c", &a, &b, &c);
    printf("age is %d, percentage is %.2f, division is %c", a, b, c);
    return 0;
}
```

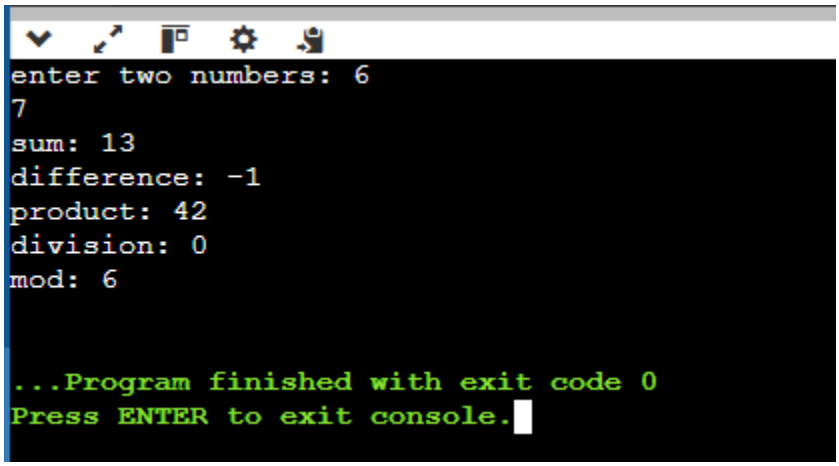


```
enter age, percentage and division: 17
89.9
r
age is 17, percentage is 89.90, division is r

...Program finished with exit code 0
Press ENTER to exit console.
```

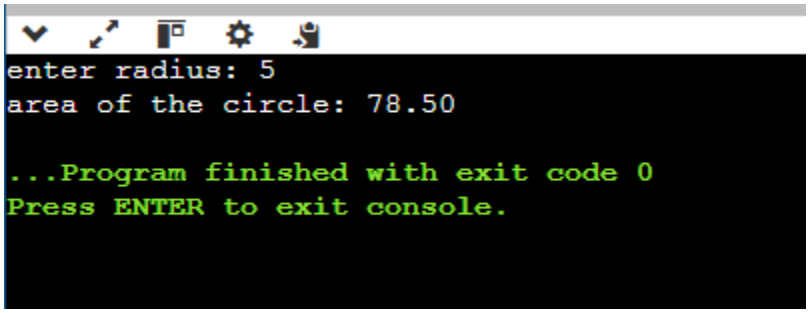
Task 2 : Arithmetic Operators

```c
#include <stdio.h>
int main() {
    int a, b;
    printf("enter two numbers: ");
    scanf("%d%d", &a, &b);
    printf("sum: %d\n", a+b);
    printf("difference: %d\n", a-b);
    printf("product: %d\n", a*b);
    printf("division: %d\n", a/b);
    printf("mod: %d\n", a%b);
    return 0;
}
```



Task 3 : Constants and Type Conversion

```c
#include <stdio.h>
#define pi 3.14
int main () {
    float r;
    printf("enter radius: ");
    scanf("%f", &r);
    printf("area of the circle: %.2f", pi*r*r);
    return 0;
}
```

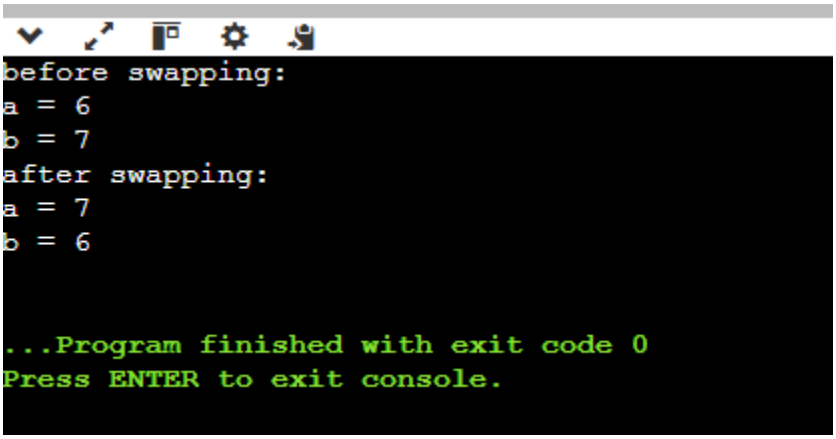


Additional Practice Tasks

1. Write a C program to swap the values of two variables using a third (temporary) variable.

```c
#include <stdio.h>

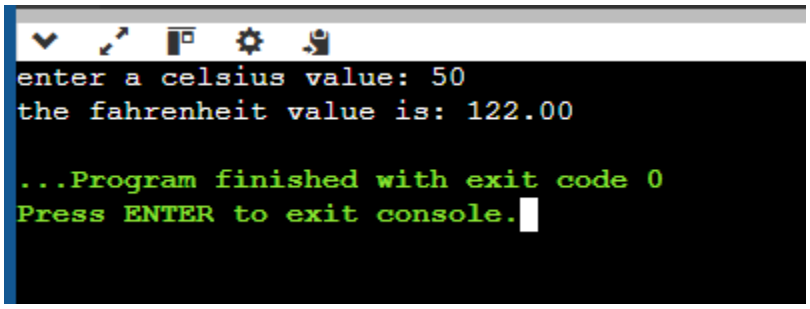
int main () {
    int a, temp, b;
    a = 6;
    b = 7;
    printf("before swapping: \na = %d\nb = %d\n", a, b);
    temp = a;
    a = b;
    b = temp;
    printf("after swapping: \na = %d\nb = %d\n", a, b);
    return 0;
}
```



2. Write a C program to convert a temperature value from Celsius to Fahrenheit (F = (C × 9/5) + 32).

```c
#include <stdio.h>

int main () {
    float c;
    printf("enter a celsius value: ");
    scanf("%f", &c);
    printf("the fahrenheit value is: %.2f", ((c*9/5)+32));
    return 0;
}
```



Write a C program to calculate simple interest, given principal, rate, and time, using a constant where appropriate.

```c
#include <stdio.h>
#define principal 50000.0
#define rate 5.5

int main() {
    float n, a, b;

    printf("enter number of years: \n");
    if (scanf("%f", &n) != 1) {
        printf("invalid\n");
        return 1;
    }
    a = (principal * rate * n) / 100.0;

    b = principal + a;

    printf("interest earned: %.2f\n", a);
    printf("total amount at the end: %.2f\n", b);

    return 0;
}
```

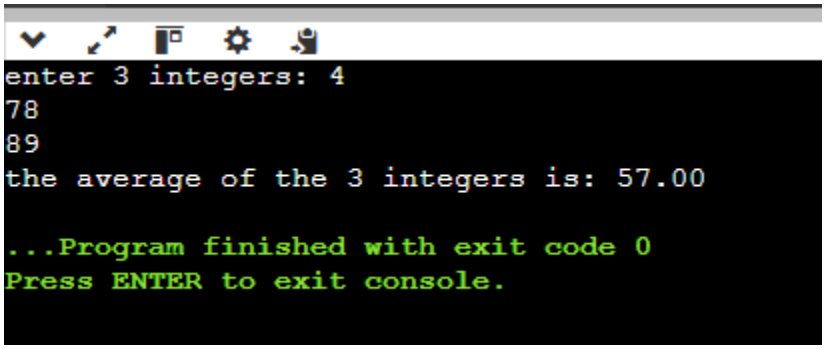
```
enter number of years:
2
interest earned: 5500.00
total amount at the end: 55500.00


...Program finished with exit code 0
Press ENTER to exit console.
```

Write a C program to find the average of three integers, expressed as a float — use explicit type casting to avoid integer division truncation.

```c
#include <stdio.h>

int main () {
    int a, b, c;
    float d;
    printf("enter 3 integers: ");
    scanf("%d%d%d", &a, &b, &c);
    d = (a+b+c)/3;
    printf("the average of the 3 integers is: %.2f", d);
    return 0;
}
```



```
enter 3 integers: 4
78
89
the average of the 3 integers is: 57.00

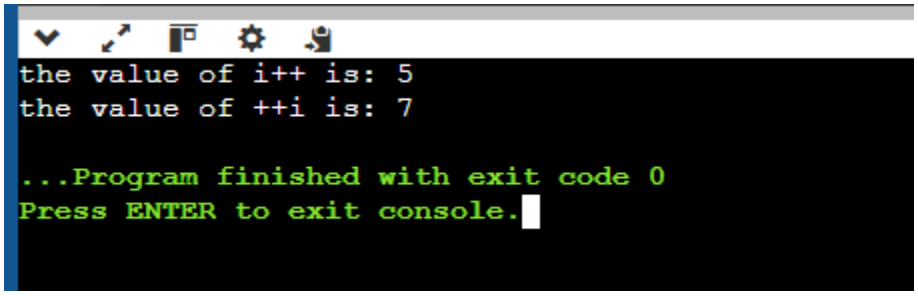
...Program finished with exit code 0
Press ENTER to exit console.
```

Write a C program to demonstrate the difference between i++ and ++i by printing the results of both in separate printf() statements.

```c
#include <stdio.h>

int main () {
    int i = 5;
    printf("the value of i++ is: %d\n", i++);
    printf("the value of ++i is: %d", ++i);
    return 0;

}
```



Ex. 1

Q.1)
Compiler: Translates the entire program into machine code at once.
Interpreter: Translates and executes code line by line.

Q.2) Data type declaration tells C the variable's type, size and how to interpret its data.

Q.3)
Variable: Value can change
Constant: Value cannot change during execution.

Q.4) #include adds declarations/functions from a header file. stdio.h provides input/output (printf() and scanf()).

Q.5)
/ : Gives the quotient. eg) 7/2 = 3
% : Gives the remainder. eg) 7%2 = 1

Q.6) i++ vs. ++i
i++ : use value first, then increment.
++i : increment first, then use value.
eg) if i = 5 → i++ = 5
                ++i = 6

Q.7)
Implicit: C converts automatically: (float x = 5)
Explicit: Programmer converts manually: (float x = (float)5/2)

Q.8) BODMAS determines operator precedence. It determines which operators are evaluated first. 10 + 2 * 3 = 16.

Q.9)
printf(): displays output
scanf(): takes user input.

Q.10) Integer division produces an integer, so the decimal part is discarded. To avoid it, use a float.